

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

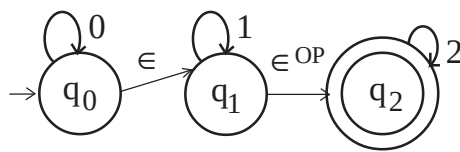
Roll No

CS-505 (GS)
B.E. V Semester Examination, June 2020
Grading System (GS)
Theory of Computation
Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
 ii) All questions carry equal marks.

1. Determine the DFA equivalent to the given NFA

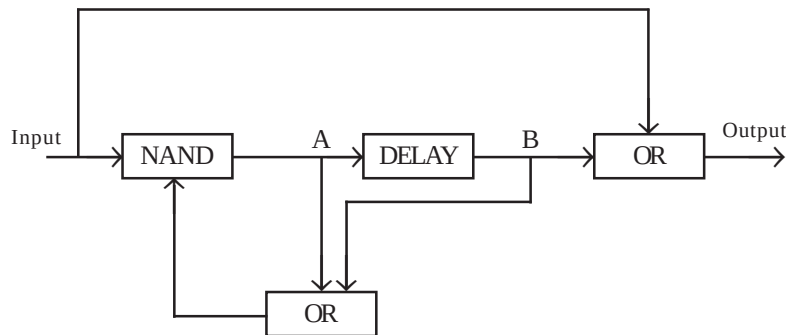


2. Convert CFG into GNF

$$S \rightarrow AA|q$$

$$A \rightarrow SS|b$$

3. What do you mean by Automata with output capability? Draw a Mealy machine equivalent to the following circuit.



4. For the following transition table, find out the string 101101 is accepted or not.

States	Input	
	0	1
$\rightarrow q_0$	q_2	q_1
q_1	q_3	q_0
q_2	q_0	q_3
q_3	q_1	q_2

5. Present a Turing machine that inserts symbol # in the beginning of a string on the Turing tape. Assume $\Sigma = \{a, b\}$.

[2]

6. Show that $L = \{a^n b^n c^n \mid n \geq 1\}$ is not CFL.

OR

Construct PDA A accepting set of all strings over $\{a, b\}^*$ with equal number of a's and b's.

7. Write short notes on:

- i) NP-Completeness
- ii) NP-Hard problems

8. Answer any two of the following:

- a) State and prove Myhill-Nerode theorem.
- b) Construct CFG to accept signed integer.
- c) State pumping lemma for context - free language.
- d) What do you mean by Turing machine? Explain multiple types Turing Machine.
